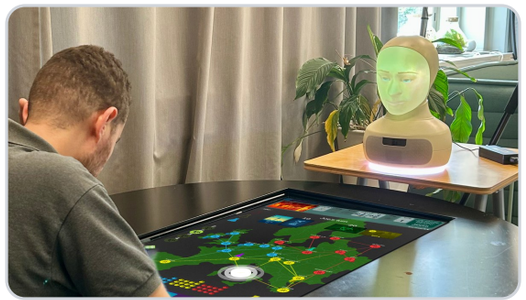
## 1 Testbed



state ↓ ↑ moves

**rule-based solver**

**HRI corpus**
70 sessions · 36 participants
speech · video · gaze · game logs

## 2 Partner signals

*training time only*

### a Self-reported

enjoyment ●●●●○
*rated once, after each game*

speech · text → **Speech-to-Joy**

### b Measured

brain activity
*fMRI, only in the scanner*

conversation → **brain encoder**

## 3 Model adaptation *current/future work*

trained

full-duplex

partner speech
partner gaze
task state
**LLM**
$h_t$
text head
gaze head
action head
bias
robot speech
robot gaze
robot action

**stage 1 · ground**
$\pi \cdot v$ self-play
conversational gaze
***general*** · *all heads*
*fine-tuned on abundant data*

**stage 2 · steer**
recurrent
intervention head
***partner-specific***
80 ms

observed
predictors